

Gurutva Patle

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RESEARCH INTERESTS

3D Computer Vision, Neural Scene Representations, Novel View Synthesis, 3D Gaussian Splatting, Surface Reconstruction, Neural Rendering, Computer Graphics, Machine Learning

EDUCATION

Indian Institute of Science (IISc), Bangalore Aug 2023 – Present

PhD, Electrical Communication Engineering – GPA: 8.20/10

Advisor: Dr. Rajiv Soundararajan

Yeshwantrao Chavan College of Engineering (YCCE), Nagpur Aug 2017 – May 2021

Bachelor of Engineering, Computer Technology – GPA: 9.02/10

Hadas Junior College, Nagpur Aug 2015 – May 2017

Class XII – Percentage : 84.77%

Ira International School, Nagpur May 2015

Class X – GPA: 10/10

PHD RESEARCH

Visual Information Processing Lab, IISc Bangalore 2023 – Present

Understanding 3D scene representation learning from sparse multi-view inputs, with a focus on 3D Gaussian Splatting, feed-forward models, surface reconstruction, and neural rendering for static and dynamic scenes.

PUBLICATIONS

AD-GS: Alternating Densification for Sparse-Input 3D Gaussian Splatting

Gurutva Patle, Nilay Girgaonkar, Nagabhushan Somraj, Rajiv Soundararajan

ACM SIGGRAPH Asia 2025

Project page: <https://gurutvapatile.github.io/publications/2025/ADGS.html>

On Sampling and Sparsity in View Synthesis

Gurutva Patle, Nilay Girgaonkar, Aayush Bansal, Rajiv Soundararajan

(Under Review)

Update-GS: Efficient 3D Gaussian Splatting Updates via Feed-Forward Initialization

Gurutva Patle, Kartikey Singh, Rajiv Soundararajan

(Under Review)

ONGOING PROJECTS

Character Motion Signature

Adobe Research

Pose-Free Sparse-View 3D Gaussian Splatting

Qualcomm Innovation Fellowship (QIF) 2026

Sparse-Input Surface Reconstruction

EXPERIENCE

Adobe Research, Research Intern May 2026 – Present

Character Motion Signature – Research on video generation and generative AI.

Mentors: Kuldeep Kulkarni, Duygu Ceylan, Balaji Vasan Srinivasan

Indian Institute of Science (IISc), Teaching Assistant Aug 2025 – May 2026

Digital Image Processing, Advanced Image Processing

Infosys Limited, Digital Specialist Engineer Aug 2021 – Jun 2023

TantraByte.Pvt.Ltd, Artificial Intelligence Intern Dec 2020 – May 2021

RELATED COURSES

Linear Algebra, Probability, Optimization, Machine Learning, Computer Vision, Image Processing, Digital Video Processing, 3D Vision and Inverse Graphics

TECHNICAL SKILLS

Programming: Python, C++, PyTorch, CUDA

Vision/Graphics: OpenCV, PyTorch3D, COLMAP, 3D Gaussian Splatting, Blender

Tools: Git, Linux, LaTeX

AWARDS

Qualcomm Innovation Fellowship (2026) | Google KIAC Fellowship (2026)

Manthan 2021 Winner – Government of India | Imagine Virtual Hackathon 2020 – Finalist (Top 8 India)

ACM ICPC 2019 – International Rank 252